Name			
Student ID			
Assessment Venue		Seat Number	

REPUBLIC POLYTECHNIC

E211 Operations Planning II

AY2021 Semester 2 End-Semester Examination (ESE)

Instructions to student:

1) For this question paper, there are **41** pages (including this cover page).
2) Note that textbooks are not allowed. You may refer to paper-based notes (in bound form, and no larger than A4 size) during the assessment.

This segment is to be used by the invigilator only and for 'online' and 'online and paper' mode assessments only.		
Please tick the box below if the student has done part of the assessment online:	Invigilator's Name:	Invigilator's Signature:
☐ Partially done online		

This Segment is to be used by staff grader(s) only.		
Question Number	**Marks Awarded**	**Max Marks**
Section A/Q1		2
Section A/Q2		2
Section A/Q3		2
Section A/Q4		2
Section A/Q5		2
Section B/Q6		20
Section B/Q7		19
Section B/Q8		20
Section B/Q9		12
Section B/Q10		13
Section B/Q11		6
Total		100

Full Marks: 100

Section A (5 Questions) **Maximum Marks for This Section: 10**

Question 1 (2 marks)

Alex wants to forecast the demand for a potential new product. He arranges a few sessions where people with different expertise are brought together for discussions. This form of forecasting is known as:

(A) Customer survey

(B) Focus group

(C) Delphi method

(D) Grassroots forecasting

Answer: ( )

Question 2 (2 marks)

Refer to the regression report in Figure 1a. Which of the following statements is **TRUE**?

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.494067764
R Square	0.244102955
Adjusted R Square	0.202108675
Standard Error	75.67585869
Observations	20

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	438.3894737	35.15381683	12.47060812	2.71001E-10
X Variable 1	-7.07518797	2.93458393	-2.41096801	0.026815897

Figure 1a: Regression Output from Excel

(A) There is strong evidence to suggest that the intercept coefficient is equal to zero.

(B) 49.4% of the variation in Y variable can be explained by the variation in X variable.

(C) As X variable increases, Y variable will decrease.

(D) The correlation between X variable and Y variable is fairly high.

Answer: ( )

Question 3 (2 marks)

Given the network with travel distances in Table 1a, the Nearest Neighbour method with **starting node at C** would give a routing solution of:

Table 1a: Travel Distance between Nodes

	A	B	C	D
A	-	32	16	23
B	32	-	25	40
C	16	25	-	27
D	23	40	27	-

(A) A – C – B – D – A

(B) C – A – D – B

(C) C – B – A – D – C

(D) C – A – D – B – C

Answer: ( )

Question 4 (2 marks)

Which queuing model best describes the following queuing system?
Patients arrive at the specialist clinic according to the time of their scheduled appointment. The consultant at the clinic spends varying amounts of time with each patient depending on patient condition.

(A) M/D/1

(B) D/M/1

(C) M/M/1

(D) D/G/2

Answer: ( )

Question 5 (2 marks)

Which of the following statements regarding simulation is **FALSE**?

(A) To validate a FlexSim model, run the simulation multiple times to ensure there is no bug.

(B) One purpose of performing simulation is to gain insights in 'what-if' situations.

(C) One advantage of simulation is that it enables long processes to be examined under a compressed timeline.

(D) Model conceptualization can be carried out using flow charts and system diagrams.

Answer: ( )

END OF SECTION A

Section B (6 Questions) **Maximum Marks for This Section: 90**

Question 6 (20 marks)

Show your workings clearly and round off your answers to two decimal places where necessary.

Aida is the product sales manager of a brand of handbag that has seen decreasing popularity. She is reviewing the quarterly sales figures (every 3 months) over the past 3 years. Figure 2a below shows the quarterly sales revenue of the handbag.

Figure 2a: Product Quarterly Sales

Question 6a (2 marks)

Can Aida use Holt's Method to forecast the sales for the next 4 quarters? Explain.

Question 6b (2 marks)

In order to remove the seasonal variation from the data, how many periods of moving average should Aida use? Give a reason.

Question 6c (6 marks)

Refer to the partial forecasting workings in Table 2a.

Table 2a: Partial Workings of Forecasting

Year/ Quarter	Actual Value (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio
2020/ 1	3200	5	6865.50		
2020/ 2	4015	6	6610.25		
2020/ 3	14875	7	6400.50	?	?
2020/ 4	4800	8	6215.15		

Question 6c(i) (2 marks)

Calculate the Centred Moving Average for **2020 Quarter 3**.

Question 6c(ii) (2 marks)

Calculate the Seasonal Ratio for **2020 Quarter 3**.

Question 6c(iii) (2 marks)

Given that the Seasonal Index for Quarter 3 is 2.32, compute the Deseasonalized Value for **2020 Quarter 3**.

Question 6d (2 marks)

Given that the linear trend in the deseasonalized values can be represented by a regression model: $y = 9702.21 - 414.85x$, compute the sales forecast for **2022 Quarter 3**.

Question 6e (2 marks)

Calculate the Absolute Percent Error for **2020 Quarter 3** using the data in Table 2b below.

Table 2b: Actual Sales and Forecast

Year/ Quarter	Actual Value (Y)	Seasonalized Forecast
2020/ 1	3200	2972.30
2020/ 2	4015	4206.42
2020/ 3	14875	15771.26
2020/ 4	4800	4721.77

Question 6f (6 marks)

Aida's company recently launched a new model of handbag. Table 2c shows the sales in the initial months.

Table 2c: New Handbag Initial Sales

Month	Sales
1	320
2	405
3	475
4	680

Question 6f(i) (2 marks)

Forecast the sales in Month 5 using 3-period Weighted Moving Average (with decreasing weights of 3, 2 and 1 from most recent to least recent data).

Question 6f(ii) (2 marks)

Forecast the sales in Month 5 using Simple Exponential Smoothing. Assume smoothing constant alpha = 0.6 and the forecast in Month 4 is 440.2.

Question 6f(iii) (2 marks)

Should Aida use Weighted Moving Average or Simple Exponential Smoothing to forecast sales of the new handbag? Give your reason.

Question 7 (19 marks)

Show your workings clearly.

A regional distribution network is shown in Figure 3a. The nodes represent cities and numbers on the arcs represent time to travel (in hours) between the cities.

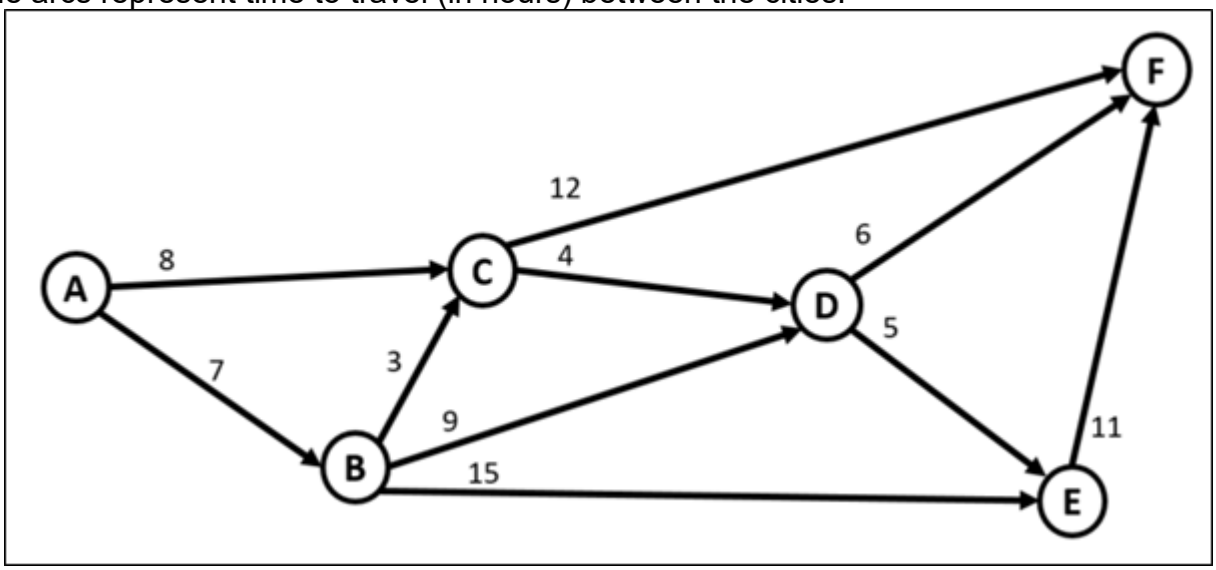

Figure 3a: Distribution Network Diagram

Question 7a (6 marks)

Table 3a below shows partial workings used in Dijkstra's Algorithm to find the shortest path from node A to each of the other nodes in the diagram in Figure 3a.

Table 3a: Partial Dijkstra's Algorithm Workings

Iteration	Solution Set	Arcs	Total Time (hrs)
Iteration 1	A	A → B A → C	7 8
Iteration 2	A B (7)	A → C B → C B → D B → E	8 **W** 16 22
Iteration 3	A B (7) C (8)	B → D B → E C → D C → F	16 22 12 20
Iteration 4	…	…	…
…	…	…	…

Question 7a(i) (2 marks)

Calculate the value of **W** in Table 3a.

Question 7a(ii) (2 marks)

Which node should be added to the solution set in **Iteration 4**? Explain.

Question 7a(iii) (2 marks)

Determine the shortest path from A to E and state the total travel time.

Question 7b (6 marks)

A Minimum Cost Flow model is used to study the distribution network in Figure 3a.
Given that:

- Node A and C can supply up to 400 and 200 units respectively
- Node E and F have demand of 250 and 300 respectively
- Nodes B and D are pure transhipment nodes

Question 7b(i) (2 marks)

Define the decision variable Xij for the Minimum Cost Flow model.

Question 7b(ii) (2 marks)

Formulate the net flow constraint for **Node D** based on the decision variables defined.

Question 7b(iii) (2 marks)

Formulate the net flow constraint for **Node A** based on the decision variables defined.

Question 7c (7 marks)

At one of the cities, a product has to be distributed to a number of stores. Table 3b shows the travel time between stores. Warehouse is at node P.

Table 3b: Travel Time Between Stores

From	To (Travel time in minutes)				
	P	**Q**	**R**	**S**	**T**
P	-	34	28	26	29
Q	34	-	41	31	38
R	28	41	-	44	39
S	26	31	44	-	40
T	29	38	39	40	-

Clarke and Wright savings method is used to plan a delivery route. Table 3c shows the **savings** between pairs of stores ranked in descending order.

Table 3c: Savings Ranked in Descending Order

Rank	Savings	From	To
1	29	Q	S
2	25	Q	T
3	**?**	Q	R
4	18	R	T
5	15	S	T
6	10	R	S

Question 7c(i) (2 marks)

Calculate the Savings between node Q and node R.

Question 7c(ii) (2 marks)

Based on the Savings Table 3c, the first sub-tour to form will be P-Q-S-P. What action should be performed at the next step (Rank 2)?

Question 7c(iii) (3 marks)

State the complete route based on Clarke and Wrights Savings Method and calculate the total travel time.

Question 8 (20 marks)

Show your workings clearly and round off your answers to three decimal places where necessary. Use ^ symbol to represent the power notation.

June runs a self-service printing shop. Walk-in customers to her shop arrive according to Poisson distribution. On average, the time between customer arrivals is 15 minutes. The service time follows negative exponential distribution.

Question 8a (2 marks)

Calculate the average customer arrival rate per hour, λ.

Question 8b (1 mark)

Given that there is 1 printer in the shop, state the appropriate queuing model (using Kendall's notation) to study the queuing process.

Question 8c (2 marks)

Given that the average service rate, μ is 5.5 per hour, calculate the utilization, ρ of the printer.

Question 8d (2 marks)

Calculate the average waiting time of customers, Wq.

Question 8e (4 marks)

With the implementation of a rapid data transfer interface, June observes that the variation of the service time has been reduced. The standard deviation of the service time is now 5 minutes.

Question 8e(i) (2 marks)

Explain why the service times should not follow negative exponential distribution and identify a suitable queuing model using Kendall's Notation.

Question 8e(ii) (2 marks)

Calculate the average customer waiting time, Wq for this queuing model.

Question 8f (7 marks)

June's printing business has grown and she plans to add more printers. Customer arrival rate has increased to 12 per hour while service rate remains at 5.5 per hour.

Question 8f(i) (3 marks)

What is the minimum number of printers June would need to have? Explain clearly.

Question 8f(ii) (2 marks)

Assume that June has sufficient number of printers set up. Given that the average number of customers in queue, Lq is 1.421, calculate the average number of customers in her shop.

Question 8f(iii) (2 marks)

June's shop has limited space for customers to wait. Assume that she has **3 printers** in the shop.
Using Table 4a where n is the number of customers in system, calculate the probability that there will be more than 1 customer in the queue.

Table 4a: M/M/k Model Partial Workings

n	$\frac{(\lambda/\mu)^n}{n!}$	$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.084	0.378
1	2.182	0.183	0.275
2	2.380	0.200	0.200
3	1.731	0.145	0.145
4	0.944	0.079	0.106
5	0.412	0.035	0.077
6	0.150	0.013	0.056

Question 8g (2 marks)

June discovers that some customers would leave the queue when they became impatient after waiting for some time. What do you call this behaviour? Can June still use the above queuing models if many customers exhibit this behaviour?

Question 9 (12 marks)

John is studying a material handling robot system at a facility using simulation. A process flow model was built in FlexSim. Figure 5a shows a snapshot of part of the model during simulation.

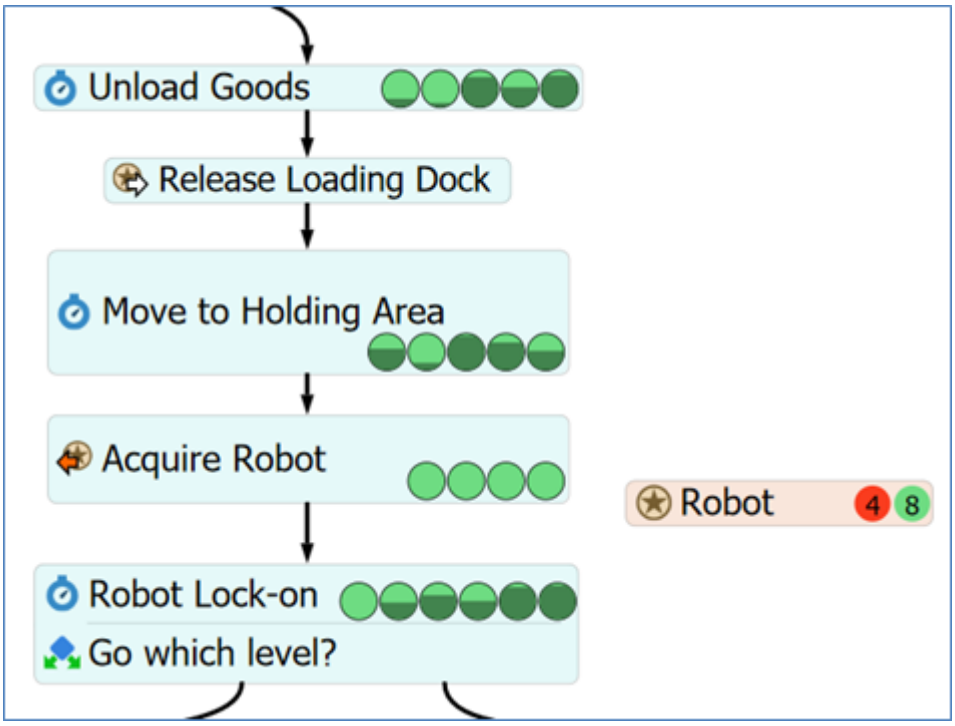

Figure 5a: FlexSim Process Flow Simulation Snapshot

Question 9a (2 marks)

From Figure 5a, state the following:

Number of Robots deployed = _____

Number of items waiting for Robot = _____

Question 9b (2 marks)

What type of activity is ‘Go which level’? What is its function here?

Question 9c (2 marks)

State **two** performance measures that John can use to study the process.

Question 9d (2 marks)

A new improved Robot lock-on mechanism has been designed. As the mechanism is totally new, John does not have any data on the mechanism timing. What can he do to model the variability of the timings?

Question 9e (4 marks)

Figure 5b shows the statistics page of the Robot resource. Answer the following questions:

Figure 5b: Robot Resource Statistics

Question 9e(i) (1 mark)

Average waiting time for a Robot = _____

Question 9e(ii) (1 mark)

Amount of space that should be provisioned for items waiting for the Robot (number of items) = _____

Question 9e(iii) (2 marks)

Calculate the Robots' average utilization.

Question 10 (13 marks)

Jane works as a process leader in the quality control (QC) department of a production line. The process consists of two parallel stations. Each station is to process a different part and is manned by an Operator. Jane has built a simulation model using FlexSim to study the process (Figure 6a).

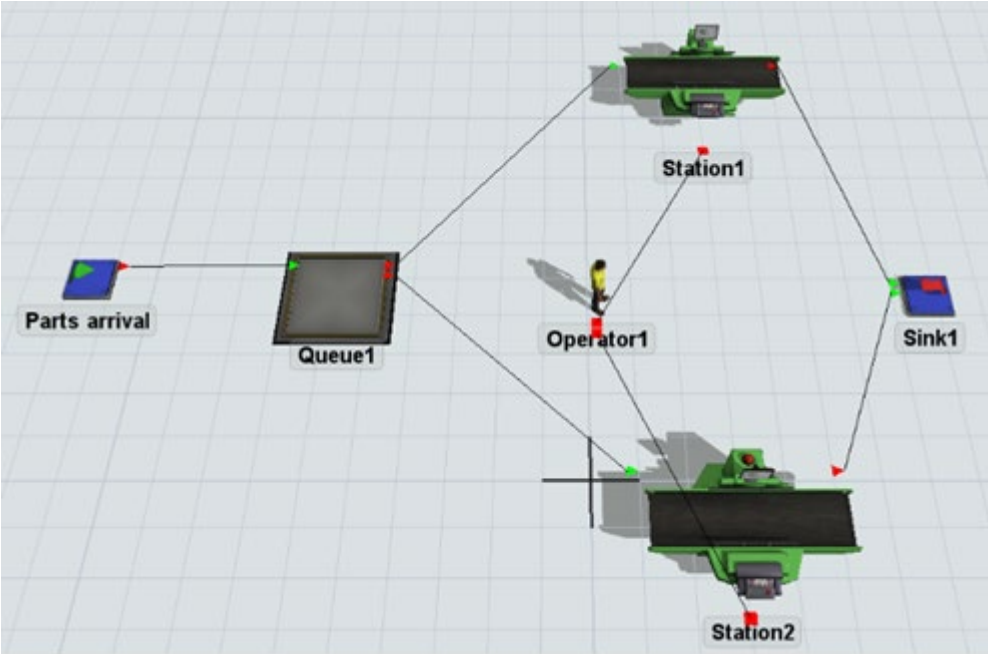

Figure 6a: Simulation Model of QC Process

Question 10a (2 marks)

Output from production is sent in batches to the stations at specific times of the day. Describe how Jane should model the arrivals based on properties page shown in Figure 6b.

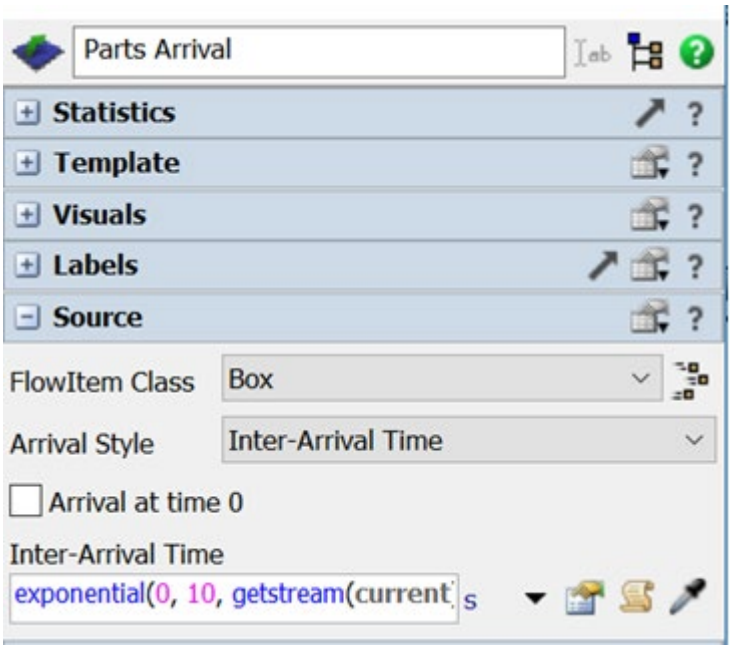

Figure 6b: Source Properties Page

Question 10b (3 marks)

In order to simulate different parts inspected at each station, Jane wants to tag the incoming parts with a label name called 'PartType'. Items with label value of 1 should be inspected at Station2 whereas items with label value of 2 should be inspected at Station1. Fill in the correct inputs for 'A', 'B' and 'C' for Figure 6c.

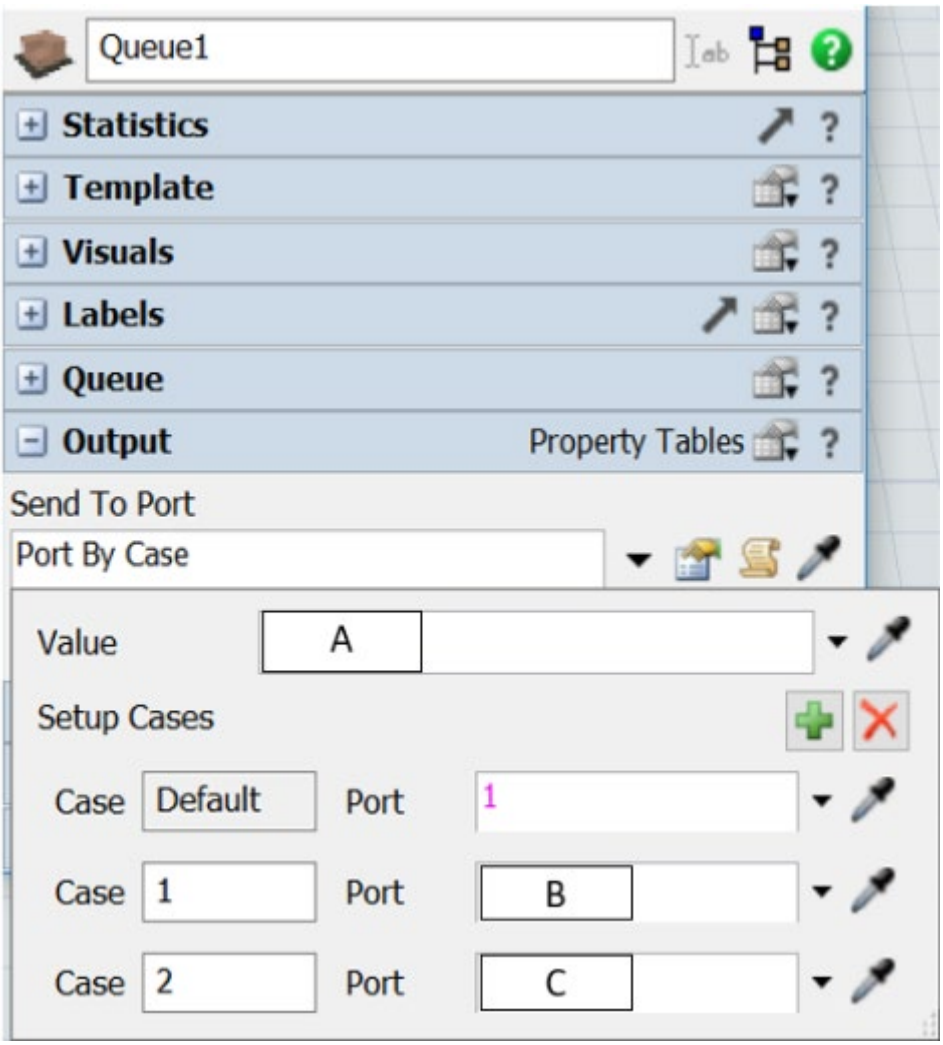

Figure 6c: Modeling the Flow from Queue to Station

A = _____

B = _____

C = _____

Question 10c (2 marks)

Referring to Figure 6a, Jane forgot to model the transport of the parts by the Operator from Queue1 to the respective stations. Describe how she should model this in FlexSim

Question 10d (2 marks)

At the end of the simulation, statistics were generated in a Dashboard (Figure 6d).

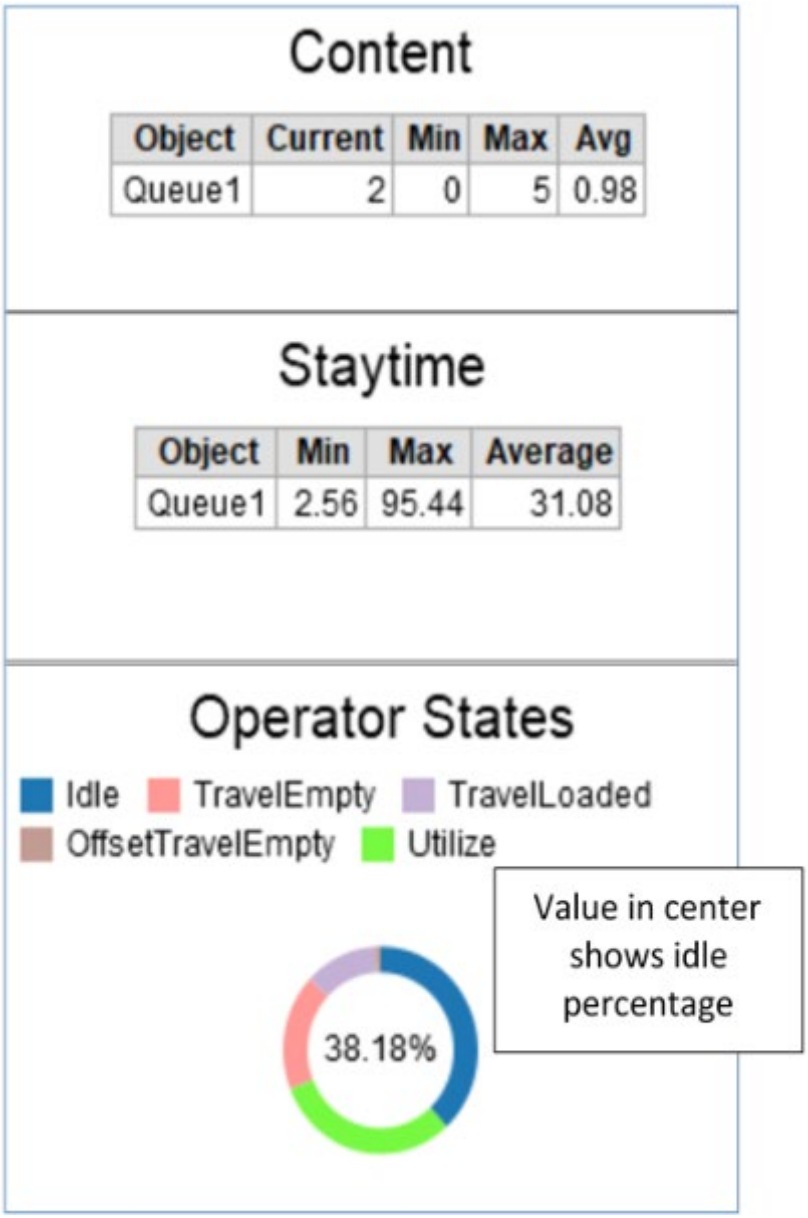

Figure 6d: Dashboard Statistics

State the following performance measures:

Average time item waits for Operator = _____

Percentage of time Operator is busy = _____

Question 10e (2 marks)

Give **two suggestions** to reduce waiting time in the queue. It is not feasible to add another operator.

Question 10f (2 marks)

Jane wants to model realistic movement paths for the Operator, taking into account the facility layout and barriers in place. How should Jane model this?

Question 11 (6 marks)

Peter constructed the Patient Flow in FlexSim Healthcare shown in following Figure 7a to simulate a registration consultation process of a clinic.

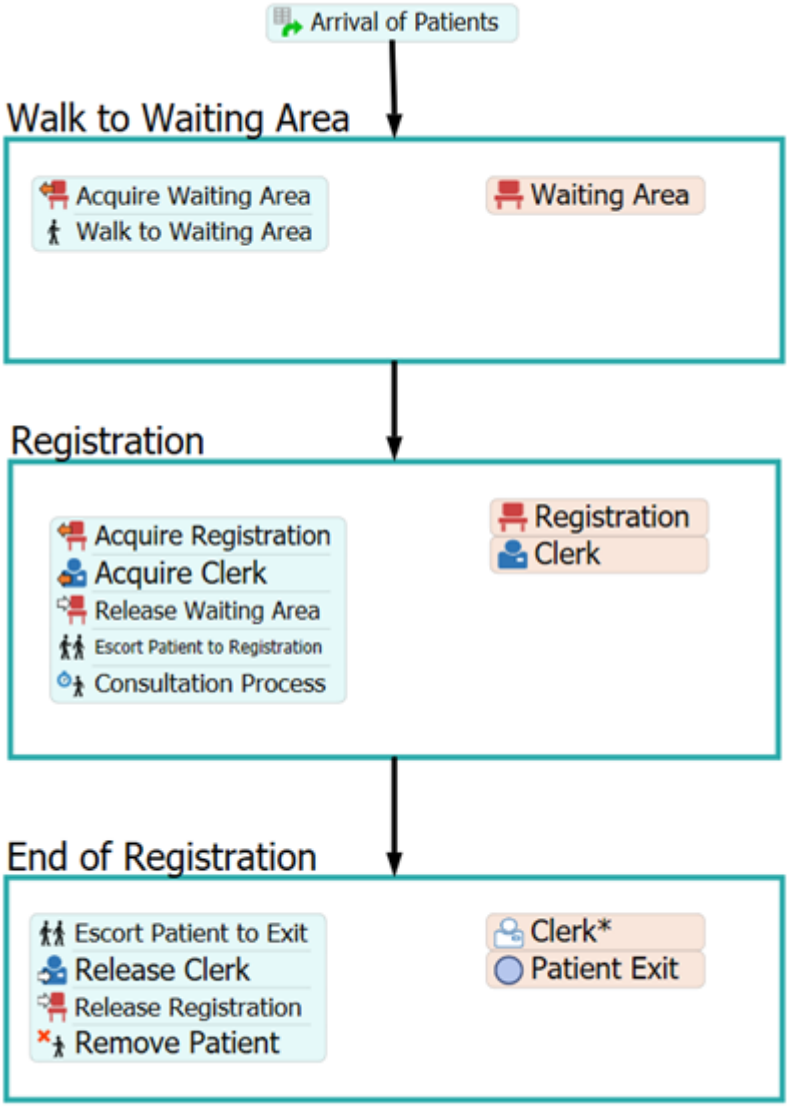

Figure 7a: Patient Flow Model in FlexSim Healthcare

Question 11a (2 marks)

According to the Patient Flow Model, what are two main roles of the Clerk?

Question 11b (2 marks)

John, the last patient of the day to arrive at the clinic, notices that there is no patient at the Registration counter, but he still has to wait at the Waiting Area. Explain.

Question 11c (2 marks)

Peter plans to use the Milestone-Milestone Time function found in the Library of Dashboard under the People Tab to measure the average time that patients need to wait at the waiting area before being served. Where should Peter insert the Milestones within the Patient Flow model created in Figure 7a? (Milestone can be found in the Library of Healthcare Process Flow as shown in Figure 7b)

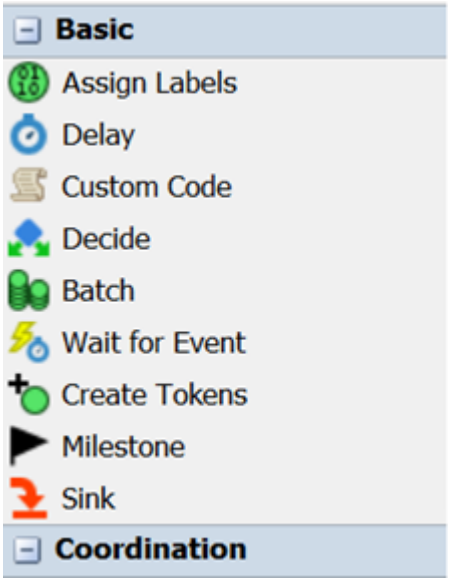

Figure 7b: FlexSim Healthcare Process Flow Basic Tools

END OF SECTION B

End of Assessment